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PROJECT REPORT

Crowdsourced Civic Issue Reporting and Resolution System

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ABSTRACT

Civic and environmental issues such as garbage accumulation, potholes, blocked drainage, and non-functional streetlights are common in urban and semi-urban areas. Traditionally, citizens report such problems through phone calls, in-person visits to municipal offices, or informal channels, resulting in poor tracking, delayed action, and little accountability on the part of civic authorities.

This project presents the design and development of a Crowdsourced Civic Issue Reporting and Resolution System — a web-based platform that allows registered citizens to report civic issues by submitting a description, a photograph, and the precise geographic location of the problem. Municipal authorities can log in to a dedicated dashboard to view incoming complaints, assign them to the relevant department, update the resolution status, and upload proof of resolution once the issue is fixed. An integrated analytics dashboard allows authorities to monitor complaint volumes, resolution times, and problem hotspots across the city.

The system is built using Python with the Flask/Django web framework for the backend, HTML, CSS, and JavaScript for the frontend, MySQL or Firebase for data storage, and the Maps API for location capture and visualization. The resulting platform improves transparency, accelerates issue resolution, and provides data-driven insights that support better municipal planning and governance.

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CHAPTER 1: INTRODUCTION

1.1 Overview

Urban local bodies are responsible for maintaining civic infrastructure such as roads, drainage, street lighting, and waste management. In practice, the sheer scale of a city makes it impossible for municipal staff to monitor every street continuously, so most civic issues are first noticed by ordinary citizens. However, the channels traditionally available to report such issues — phone calls, physical visits, or complaint registers — are slow, poorly tracked, and offer no visibility into what happens after a complaint is lodged.

The Crowdsourced Civic Issue Reporting and Resolution System addresses this gap by giving citizens a simple digital channel to report problems with photographic and location evidence, while giving municipal authorities a structured workflow to triage, assign, resolve, and analyze these reports. By digitizing the entire complaint lifecycle, the platform benefits both citizens, who gain transparency and faster resolutions, and authorities, who gain accountability and data to plan civic maintenance more effectively.

1.2 Problem Statement

Existing complaint-handling processes used by many municipal bodies suffer from the following shortcomings:

- Complaints are scattered across phone calls, letters, and informal channels with no centralized digital record.
- Citizens have no way to attach photographic or location evidence, making verification and prioritization difficult.
- There is no mechanism for citizens to track the status of a submitted complaint.
- Complaints are not systematically routed to the department responsible for resolving them.
- Authorities lack aggregated data on complaint volumes, categories, or geographic hotspots, which limits data-driven planning.

These gaps result in delayed resolutions, reduced citizen trust in civic authorities, and inefficient allocation of municipal resources.

1.3 Objectives

The primary objectives of this project are as follows:

1. To allow citizens to register and log in securely to the platform.
2. To allow citizens to report civic issues (garbage, potholes, drainage, streetlights, and similar categories) with a description, photograph, and geotagged location.
3. To allow municipal authorities to view, filter, and assign complaints to the appropriate department.
4. To allow authorities to update complaint status and upload proof of resolution.
5. To notify citizens of status changes and maintain a history of their submitted complaints.

6. To provide authorities with an analytics dashboard summarizing complaint volume, category-wise distribution, resolution time, and geographic hotspots.

1.4 Scope of the Project

The scope of this project covers the design and development of a web-based application with two primary user roles — Citizen and Authority/Admin. The system covers issue reporting, department-wise assignment, status tracking, resolution-proof verification, and analytics reporting. It does not, in its current scope, cover payment processing, integration with third-party government ERP systems, or native mobile applications, although these are identified as directions for future enhancement in Chapter 8.

CHAPTER 2: LITERATURE SURVEY

Several categories of existing systems and research were reviewed prior to designing this platform:

2.1 Manual and Call-Centre-Based Complaint Systems

Many municipal bodies still rely on manual registers or call centres to log citizen complaints. While functional, these systems depend heavily on manual data entry, are prone to human error, and provide no automated tracking or analytics. Citizens typically have no way to independently verify the status of their complaint without contacting the office again.

2.2 Existing E-Governance Complaint Portals

Various government e-governance portals (e.g., municipal grievance redressal portals) allow citizens to submit complaints online. However, many such systems are generic grievance systems not tailored to civic/environmental issues, often lack photo or geolocation capture, and provide limited department-level workflow management or public analytics.

2.3 Crowdsourced Reporting Applications

Globally, crowdsourced civic-reporting concepts (such as FixMyStreet-style platforms) have demonstrated that enabling citizens to report issues with photos and map locations significantly increases the volume and quality of civic issue data available to local authorities. These systems highlight the value of location-based reporting and public visibility of complaint status, both of which are incorporated into the design of this project.

2.4 Summary of Findings

The literature and existing systems reviewed indicate a consistent need for: (a) an accessible reporting interface for citizens, (b) location and photographic evidence to aid verification, (c) structured department-wise workflows for authorities, and (d) analytics for planning. This project is designed to incorporate all four elements into a single unified platform.

CHAPTER 3: SYSTEM ANALYSIS

3.1 Existing System

In the existing (manual/semi-digital) system, complaints are typically registered through phone calls or physical visits. There is no unified record, no proof of the issue, no live tracking for the citizen, and no systematic routing to departments. Consequently, resolution timelines are inconsistent and authorities have little visibility into problem trends across the city.

3.2 Proposed System

The proposed system introduces a digital, role-based platform with the following characteristics:

- Citizens can register, log in, and report issues from a web (and optionally mobile) interface at any time.
- Every complaint includes a mandatory photograph and a geotagged location captured via the Maps API.
- Complaints move through clearly defined statuses: Pending → Assigned → In Progress → Resolved.
- Authorities assign complaints to the correct department and must upload resolution proof before a complaint can be closed.
- An analytics dashboard aggregates complaint data by category, department, location, and time to support planning.

3.3 Feasibility Study

3.3.1 Technical Feasibility

The system relies on widely adopted, well-documented technologies (Python, Flask/Django, HTML/CSS/JavaScript, MySQL/Firebase, and a Maps API), all of which are freely or affordably available and well-suited to building a web application of this scale. No specialized or proprietary hardware is required.

3.3.2 Economic Feasibility

The technologies proposed are largely open-source, and cloud-hosted database options such as Firebase offer free or low-cost tiers suitable for a pilot deployment. This keeps development and initial hosting costs low relative to the operational benefits of faster issue resolution.

3.3.3 Operational Feasibility

The platform is designed with a simple, intuitive interface for citizens and a structured dashboard for authorities, requiring minimal training. Given the widespread use of smartphones and internet access, citizen adoption is expected to be practical in most urban and semi-urban settings.

3.4 Software and Hardware Requirements

Category	Requirement
Operating System	Windows / Linux / macOS (development); Linux (server)
Backend Language & Framework	Python 3.x with Flask or Django

Category	Requirement
Frontend Technologies	HTML5, CSS3, JavaScript
Database	MySQL (relational) or Firebase (NoSQL/cloud)
Mapping Service	Google Maps API / equivalent Maps API
IDE / Editor	VS Code, PyCharm, or equivalent
Version Control	Git and GitHub
Minimum Hardware (Development)	Dual-core processor, 4 GB RAM, 20 GB free storage
Server Hardware (Deployment)	Cloud VM or hosting service with adequate storage for media uploads

CHAPTER 4: SYSTEM DESIGN

4.1 System Architecture

The system follows a three-tier architecture consisting of a presentation layer, an application layer, and a data layer.

- **Presentation Layer:** The web (and optional mobile) interface built with HTML, CSS, and JavaScript, through which citizens and authorities interact with the system.
- **Application Layer:** The Python-based backend (Flask or Django) that implements business logic, authentication, complaint routing, and exposes APIs consumed by the frontend.
- **Data Layer:** MySQL or Firebase, used to persist user accounts, complaint records, department information, and resolution logs, along with associated media (photographs).

The Maps API is integrated at both the presentation layer (to capture and display locations) and the application layer (to process geographic coordinates for hotspot analysis).

4.2 Data Flow Overview

At a high level, data flows through the system as follows: a citizen submits a complaint (description, photo, location) from the presentation layer; the application layer validates and stores this data in the data layer and tags it with an initial "Pending" status; an authority retrieves pending complaints, assigns a department, and the status is updated to "Assigned" and later "In Progress"; upon completion, the authority uploads resolution proof, and the status changes to "Resolved", which is then reflected back to the citizen and factored into the analytics dashboard.

4.3 Database Design

The core entities of the database and their key attributes are summarized below.

4.3.1 Users Table

Field	Type	Description
user_id	INT (PK)	Unique identifier for each user

Field	Type	Description
name	VARCHAR	Full name of the user
email	VARCHAR	Login email address (unique)
password	VARCHAR	Hashed password
role	ENUM	'citizen' or 'authority'
phone	VARCHAR	Contact number
address	VARCHAR	Residential / office address

4.3.2 Complaints Table

Field	Type	Description
complaint_id	INT (PK)	Unique identifier for each complaint
user_id	INT (FK)	References the reporting citizen
type	VARCHAR	Garbage / Pothole / Drainage / Streetlight / Other
description	TEXT	Citizen-provided description of the issue
photo_url	VARCHAR	Path/URL to the uploaded photograph
latitude	DECIMAL	Geographic latitude of the issue
longitude	DECIMAL	Geographic longitude of the issue
status	ENUM	'Pending' / 'Assigned' / 'In Progress' / 'Resolved'
created_at	DATETIME	Timestamp of submission

4.3.3 Departments Table

Field	Type	Description
dept_id	INT (PK)	Unique identifier for each department
dept_name	VARCHAR	e.g., Sanitation, Roads, Electrical, Water
category_handled	VARCHAR	Issue category the department is responsible for
contact_info	VARCHAR	Department contact details

4.3.4 Resolution Log Table

Field	Type	Description
log_id	INT (PK)	Unique identifier for each resolution record
complaint_id	INT (FK)	References the related complaint
assigned_dept_id	INT (FK)	References the assigned department
resolution_photo	VARCHAR	Path/URL to proof-of-resolution photograph

Field	Type	Description
resolved_at	DATETIME	Timestamp of resolution

Relationships: A user (citizen) can raise many complaints (one-to-many). Each complaint has at most one resolution log entry (one-to-one). A department can appear in many resolution log entries (one-to-many).

4.4 Module Description

4.4.1 Citizen Module

- Registration and secure login
- Complaint submission with description, photo upload, and location capture
- Real-time status tracking of submitted complaints
- Notification and complaint history view

4.4.2 Authority / Admin Module

- Secure authority login
- View and filter complaints by type, location, or status
- Assign complaints to the relevant department
- Update complaint status and upload resolution proof
- Access the analytics dashboard for monitoring trends

4.5 User Interface Design (Overview)

The user interface is designed to be minimal and task-focused. The citizen-facing interface consists of a registration/login screen, a "Report an Issue" form (description, photo upload, map-based location picker), and a "My Complaints" screen showing status history. The authority-facing interface consists of a login screen, a complaints table with filters, a complaint detail view for assignment and status updates, and an analytics dashboard with charts summarizing complaint data.

CHAPTER 5: SYSTEM IMPLEMENTATION

5.1 Technology Stack

Layer	Technology
Backend Language	Python
Web Framework	Flask / Django
Frontend	HTML5, CSS3, JavaScript
Database	MySQL (relational) or Firebase (cloud/NoSQL)
Location Services	Maps API (geocoding, geotagging, hotspot visualization)
Media Storage	Local/cloud file storage for photographs

5.2 Implementation Details

5.2.1 User Authentication

User registration and login are implemented with hashed password storage (e.g., using a secure hashing algorithm such as bcrypt) and session- or token-based authentication to distinguish between citizen and authority roles, ensuring that authority-only features (assignment, status update, analytics) are protected from citizen access.

5.2.2 Complaint Submission

The complaint submission form captures the issue type, a free-text description, an uploaded photograph, and the geographic location. The location is captured either by allowing the user's browser to share its GPS coordinates or by letting the user select a point on an embedded map, using the Maps API. On submission, the backend validates the input, stores the photograph, and inserts a new record into the Complaints table with an initial status of "Pending".

5.2.3 Complaint Assignment and Status Workflow

Authorities view incoming complaints through a dashboard that supports filtering by category, location, and status. When a complaint is reviewed, the authority assigns it to the department responsible for that category of issue (for example, garbage complaints are routed to the Sanitation department). The complaint status is then updated programmatically as it moves through the workflow: Pending, Assigned, In Progress, and finally Resolved.

5.2.4 Resolution Proof and Closure

Before a complaint can be marked as "Resolved", the assigned department is required to upload a photograph demonstrating that the issue has been fixed. This resolution proof, along with the timestamp, is stored in the Resolution Log table and linked to the original complaint, creating an auditable record of the fix.

5.2.5 Analytics Dashboard

The analytics dashboard aggregates data from the Complaints and Resolution Log tables to display: complaint volume by category, complaint status breakdown (Pending / In Progress / Resolved), average resolution time, and a map-based hotspot visualization showing areas with a high concentration of complaints. These insights help authorities prioritize resource allocation and identify recurring problem areas.

CHAPTER 6: TESTING

6.1 Testing Approach

The system was tested using a combination of unit testing (verifying individual functions such as form validation and status updates), integration testing (verifying that the frontend, backend, and database

work together correctly), and user acceptance testing (verifying that the system meets the needs of both citizen and authority users).

6.2 Sample Test Cases

Test Case	Input	Expected Output	Result
Citizen registration	Valid name, email, password	Account created successfully	Pass
Duplicate registration	Already-registered email	Error: email already in use	Pass
Complaint submission without photo	Description + location, no photo	Validation error prompting photo upload	Pass
Complaint submission (valid)	Description, photo, location	Complaint stored with status "Pending"	Pass
Authority assigns department	Select department for a complaint	Status updates to "Assigned"	Pass
Resolution without proof photo	Attempt to mark "Resolved" with no photo	Validation error; status unchanged	Pass
Resolution with proof photo	Upload resolution photo	Status updates to "Resolved"; log created	Pass
Dashboard data aggregation	Multiple complaints across categories	Correct counts and charts displayed	Pass

CHAPTER 7: RESULTS AND DISCUSSION

The developed platform successfully allows citizens to register and submit civic complaints with photographic and location evidence, and allows municipal authorities to manage the full complaint lifecycle from assignment through verified resolution. Testing confirmed that the core workflows — registration, complaint submission, department assignment, status updates, and resolution-proof verification — function as intended.

The analytics dashboard was found to provide a clear, aggregated view of complaint categories, status distribution, and geographic hotspots, giving authorities data that was previously unavailable through manual complaint-handling processes. Overall, the system demonstrates a significant improvement over the existing manual process in terms of transparency, traceability, and accountability, since every complaint now has a verifiable digital trail from submission to resolution.

A few limitations were also observed during development and testing: the accuracy of the reported location depends on the user's device GPS or manual map selection, and the system currently assumes reliable internet connectivity for both citizens and authorities. These limitations, along with potential enhancements, are discussed further in Chapter 8.

CHAPTER 8: CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The Crowdsourced Civic Issue Reporting and Resolution System successfully bridges the gap between citizens and municipal authorities by replacing an ad hoc, opaque complaint-handling process with a structured, transparent, and accountable digital workflow. Citizens are empowered to report civic and environmental issues quickly and track their resolution, while authorities gain the tools to assign, resolve, and analyze complaints efficiently. The result is a platform that has the potential to improve civic responsiveness, strengthen citizen trust, and support more data-driven urban governance.

8.2 Future Scope

- AI-based auto-classification of issue type from uploaded photographs, reducing manual categorization effort.
- Multilingual interface support to widen accessibility across diverse citizen populations.
- SMS or WhatsApp-based reporting for citizens without access to smartphones or the internet.
- Gamification features, such as civic engagement points or recognition, to encourage citizen participation.
- Predictive analytics using historical complaint data to anticipate and proactively address recurring problem areas.
- Native mobile applications for Android and iOS to improve accessibility and adoption.

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